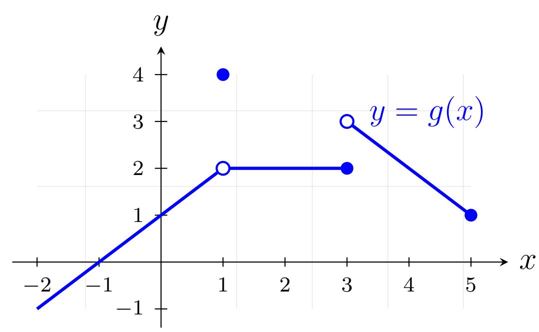
# Three Kinds of Evidence About a Limit

*Reading limits from a graph, estimating them from a table of values, and confirming them with algebra*

**Before you start.** A limit claim can be supported three ways, and a good answer says which one you used.

- **Graphical** — follow the curve in from each side and see what height it heads for. A filled dot tells you the function *value*; it does not tell you the limit.
- **Numerical** — evaluate at inputs closing in from both sides. A table suggests a limit; it never proves one.
- **Symbolic** — factor, rationalize, or clear the compound fraction, then substitute. This is the one that settles the matter.

**Graph of $g$** — used by problems 3 and 6 only. Open circles are excluded points; filled dots are included.




**1.** Evaluate $\lim_{x\to 3}\bigl(x^2 - 2x + 5\bigr)$, and state in one phrase why direct substitution is allowed here.

Answer: ________________

**2.** Let $f(x) = \dfrac{x^2 - 9}{x - 3}$. Complete the table, then say what limit the values are pointing at.

| $x$ | 2.9 | 2.999 | 3.001 | 3.1 |
|---|---|---|---|---|
| $f(x)$ | | | | |

$\lim_{x \to 3} f(x) \approx$ __________

Answer: ________________

**3.** Use the graph of $g$ above. State $\lim_{x \to 1} g(x)$ and the value $g(1)$, and name the kind of discontinuity at $x = 1$. Then, given that $g(x) = x + 1$ on the branch to the left of 1, confirm $\lim_{x \to 1^-} g(x)$ algebraically.

Answer: ________________

**4.** Evaluate $\lim_{x \to 2} \dfrac{x^2 - 4}{x^2 - x - 2}$. Substitution gives $\frac{0}{0}$, so it tells you nothing — factor first.

Answer: ____________________

**5.** Let $f(x) = \dfrac{x^3 - 8}{x - 2}$. Complete the table, state the limit it suggests, then confirm that limit algebraically.

| $x$ | 1.9 | 1.99 | 2.01 | 2.1 |
|---|---|---|---|---|
| $f(x)$ | | | | |

Answer: ____________________

**6.** Use the graph of $g$ above. Give $\lim_{x \to 3^-} g(x)$ and $\lim_{x \to 3^+} g(x)$, and explain why $\lim_{x \to 3} g(x)$ does not exist. Given that $g(x) = 6 - x$ on the branch to the right of 3, confirm the right-hand limit algebraically.

Answer: ____________________

**7.** Evaluate $\lim_{x \to 0} \dfrac{\dfrac{1}{x+2} - \dfrac{1}{2}}{x}$. Combine the two fractions in the numerator before you do anything else.

Answer: ____________________

**8.** The function $f(x) = \dfrac{x^2 + ax + 3}{x - 1}$ has a finite limit as $x \to 1$.

(a) The numerator must be zero at $x = 1$ for that to happen. Use this to find $a$.

(b) With that value of $a$, evaluate $\lim_{x \to 1} f(x)$.

Answer: ____________________

**9.** Let $f(x) = \dfrac{|x - 4|}{x - 4}$. Complete the table, then explain why $\lim_{x \to 4} f(x)$ does not exist even though both one-sided limits do.

| $x$ | 3.9 | 3.99 | 4.01 | 4.1 |
|---|---|---|---|---|
| $f(x)$ | | | | |

Answer: ____________________

**10. Synthesis.** Let $p(x) = \dfrac{x^2 + x - 6}{x - 2}$ for $x \neq 2$, with $p(2)$ not yet defined.

(a) Evaluate $p$ at $x = 1.9$ and at $x = 2.1$.

(b) Find $\lim_{x \to 2} p(x)$ algebraically, and state the value of $p(2)$ that would make $p$ continuous at 2.

(c) Sketch the graph of $p$ near $x = 2$, marking the hole, and write one or two sentences saying how the table, the algebra and your sketch agree.

Answer: ____________________